

YASH BAJPAI

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Professional Summary

Results-driven IT student with expertise in full-stack development, machine learning, and data structures. Proven track record in Python, JavaScript, and database management through internship and academic projects.

Education

Vellore Institute of Technology (VIT)

Bachelor of Technology in Information Technology, CGPA: 8.53/10

2023 – 2027

Vellore, India

Delhi Public School Vindhyanagar

Higher Secondary Education (CBSE) - PCM, 88%

2023

India

Technical Skills

Programming Languages: Python, C/C++, Java, JavaScript (ES6+), SQL, HTML5, CSS3

Web Development: Node.js, Express.js, React.js, RESTful APIs, JSON Web Tokens (JWT), Bootstrap

Database Management: MySQL, MongoDB, PostgreSQL, Database Design, Query Optimization

Machine Learning & Data Science: NumPy, SciPy, scikit-learn, OpenCV, Pandas, Matplotlib, Feature Engineering

Tools & Technologies: Git/GitHub, Docker, Postman, Jest, Nodemailer, QR Code Generation, Linux/Windows

Core Computer Science: Data Structures & Algorithms, Software Engineering, System Design, Object-Oriented Programming

Experience

Summer Intern - Algoleap

June 2025 – July 2025

Web Scraping & Data Processing

Remote

- Developed modular web scraper using Python and BeautifulSoup for company reviews from Glassdoor and AmbitionBox, processing entries with site-specific parsing logic
- Implemented sentiment-based filtering using NLP techniques to identify negative reviews and structured data output in CSV/JSON formats with 95% success rate
- Built robust error handling, logging system, and retry mechanisms with exponential backoff for production-ready data pipelines and anti-scraping countermeasures
- Designed scalable automation scripts with modular architecture enabling easy integration of new review platforms and data sources

Technical Projects

RSVP Management System | Node.js, Express.js, MongoDB, JWT, Nodemailer

2024

- Built comprehensive full-stack RSVP system with MVC architecture, featuring responsive UI, secure JWT authentication with bcrypt password hashing, and session management
- Integrated automated email confirmations via Nodemailer with customizable templates and dynamic QR code generation using node-qrcode for seamless event check-in
- Implemented real-time RSVP tracking with WebSocket connections and achieved 100% test coverage using Jest for unit testing and Postman for comprehensive API testing

Enhanced Object Detection | Python, OpenCV, NumPy, scikit-learn, COCO Dataset

2024

- Conducted binary object classification using COCO 2017 dataset with 80,000+ images, implementing bilateral filtering for edge preservation and noise reduction
- Extracted 500 keypoints using Scale-Invariant Feature Transform (SIFT) and applied Principal Component Analysis for dimensionality reduction from 500 to 50 features
- Computed Shannon entropy for high-information feature selection and trained SVM classifier with RBF kernel, improving classification accuracy from 53% to 63% through feature engineering

Sparse Image Compression using DCT | Python, NumPy, SciPy, PIL

2024

- Developed efficient image compression tool using 8x8 block-wise Discrete Cosine Transform and sparse matrix techniques, achieving 70% file size reduction while maintaining PSNR above 30dB
- Implemented frequency domain analysis with adaptive quantization tables and optimized compression algorithms for maximum efficiency while preserving visual quality